

MICROSOFT FABRIC / REAL-TIME INTELLIGENCE

When the wind turns

A wildfire is not a data problem until you have to decide who to warn first — and you have ninety seconds.

LIVE FABRIC WORKSPACE

Everything you are about to see is a running deployment — not a mock-up.

24

Fabric items

721.6K

rows ingested

12 / 12

tables in OneLake

89

dispatch events

A commander does not ask “where is the fire?”

They already know where it is.

Somebody phoned it in.

The question that actually costs lives is:

**“The wind just shifted.
Who is downwind, and can we reach them?”**

That question crosses four systems, three agencies and one very long phone call.

WHERE THE ANSWER LIVES TODAY

Satellite feed

Agency A · CSV by email

Weather model

Agency B · separate portal

Population + schools

Municipal GIS · shapefiles

Fire stations, water

Regional ops · spreadsheet

Aircraft positions

ADS-B · third-party app

Five sources. Five owners. No shared clock.

One incident. Five questions. One workspace.

We follow the commander's questions — not a tour of the product menu.



The hero of this story is the connected evidence chain — not an autonomous decision.

This is a running workspace, not a slideware diagram

24 deployed items, under source control, syncing to Git.

	Name	Status	Git status	Type	Task	Owner	Refreshed	Next refresh	Endo...
	00_config		✓ Synced	Notebook	—	System Adm...	—	—	—
	01_reference_data		✓ Synced	Notebook	—	System Adm...	—	—	—
	02_ingest_satellite_fires		✓ Synced	Notebook	—	System Adm...	—	—	—
	03_ingest_fire_weather		✓ Synced	Notebook	—	System Adm...	—	—	—
	04_ingest_aircraft		✓ Synced	Notebook	—	System Adm...	—	—	—
	06_ont_wildfires_curation		✓ Synced	Notebook	—	System Adm...	—	—	—
	07_graph_wildfire_snapshot		✓ Synced	Notebook	—	System Adm...	—	—	—
	ACT_Wildfire		✓ Synced	Activator	—	System Adm...	—	—	—
	AGENT_Wildfire		✓ Synced	Data agent	—	System Adm...	—	—	—
	EH_Wildfire		✓ Synced	Eventhouse	—	System Adm...	—	—	—
	EH_Wildfire		✓ Synced	KQL Database	—	System Adm...	—	—	—
	ES_Wildfire		✓ Synced	Eventstream	—	System Adm...	—	—	—
	GQS_Wildfire_Story		✓ Synced	Graph query...	—	System Adm...	—	—	—
	GRAPH_Wildfire_Impact		✓ Synced	Graph model	—	System Adm...	—	—	—
	LH_WildFires		✓ Synced	Lakehouse	—	System Adm...	—	—	—
	LH_WildFires		—	SQL analytic...	—	System Adm...	—	—	—

ws_wild_fires_demo · workspace item list, 8 September 2026

Eventstream, Eventhouse, Lakehouse,
Ontology, Graph, Activator, Agents

7 notebooks, all Synced to Git —
this deploys into your tenant

One workspace. One security
boundary. One bill.

**No copies. No nightly export.
No “let me send you the file.”**

The signal arrives every ten minutes

Two satellite perspectives answer two different questions. We ingest both.

GEOSTATIONARY

Meteosat · Met12 · every 10 minutes
It cannot see detail — but it sees change.
This is what makes it early warning.

POLAR

VIIRS 375 m · MODIS ~1 km · 2–4 passes/day
It cannot see change — but it sees shape.
Together: cadence plus resolution.

A polar orbiter tells you a fire exists. Only cadence tells you it is accelerating.

Before you run this notebook

Pulls active fire detections from NASA FIRMS and publishes them to `ES_Wildfire`, which routes them into `bronze_fire_raw`. Runs every 10 minutes to match the geostationary refresh cycle.

What to configure

Everything is in the **Configuration** cell, immediately below this one.

Setting	Purpose	Notes
FIRMS_KEY	NASA FIRMS map key	Required. See below
WS, ES_ID, ES_SRC	Where to publish	See the table below
WEST, SOUTH, EAST, NORTH	Area of interest	FIRMS expects west, south, east, north
TIERS	Satellite products queried	Geostationary gives cadence, polar gives resolution

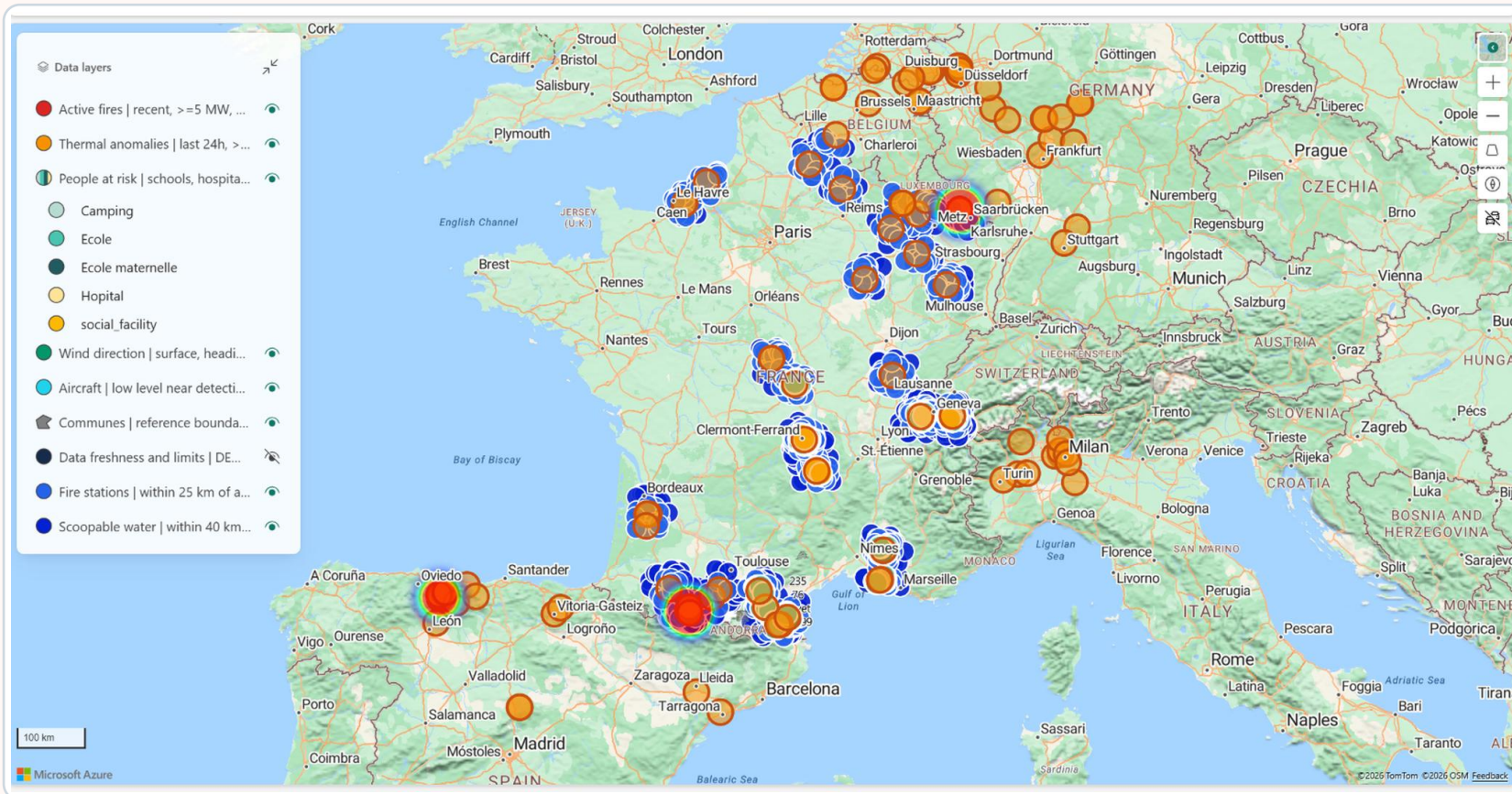
02_ingest_satellite_fires · NASA FIRMS → ES_Wildfire

WHAT THE NOTEBOOK DOES

- Queries NASA FIRMS for the Europe bounding box
- Publishes typed events to the ES_Wildfire endpoint
- Runs on a 10-minute schedule to match Met12
- Reads its key from Key Vault, not from the notebook
- One FIRMS transaction per satellite tier, per run

The map becomes the shared situation picture

Twelve live layers. Thermal observations, people at risk, wind, aircraft, stations, water.



Active fires

recent, ≥ 5 MW

People at risk

schools, hospitals, camping

Wind direction

surface, coloured by speed

Response options

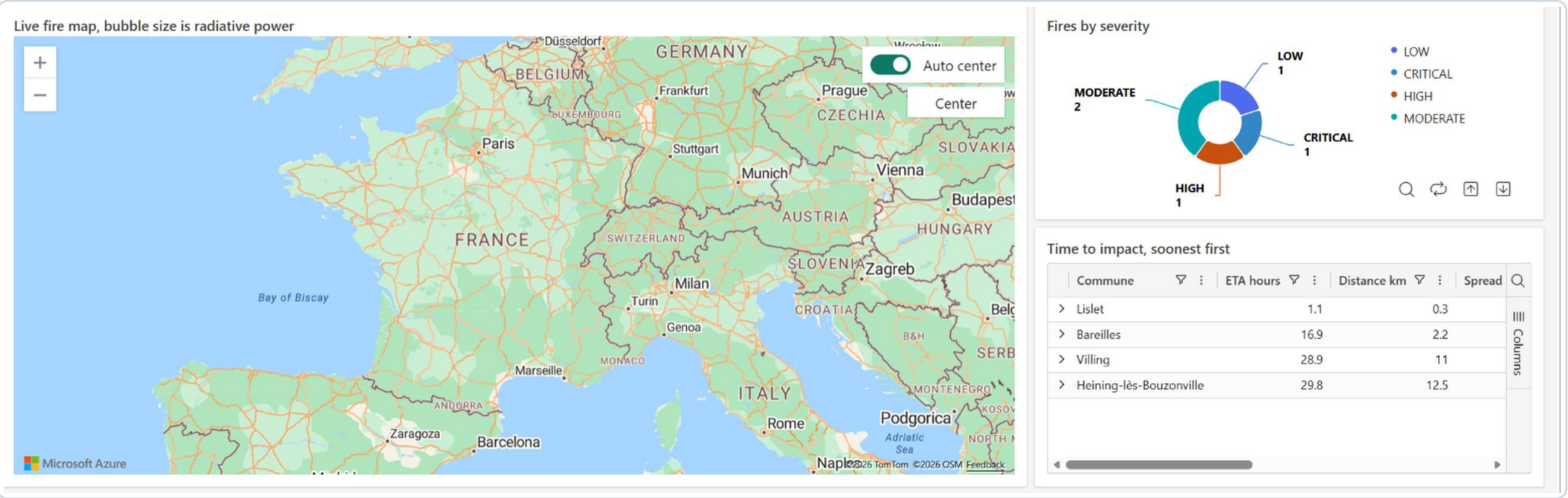
stations ≤ 25 km, water ≤ 40 km

Wildfire Atlas – Live · 12 layers, KQL + Lakehouse geography

Every layer is one KQL function.

The wind turns — and the ranking changes

“Which dot is biggest?” is the wrong question. “Who is downwind?” is the right one.



RTD_Wildfire_Command · live, 09:27 CEST 8 September 2026

4

active fire zones
right now

795.8 %

growth on the
CRITICAL zone

201,198

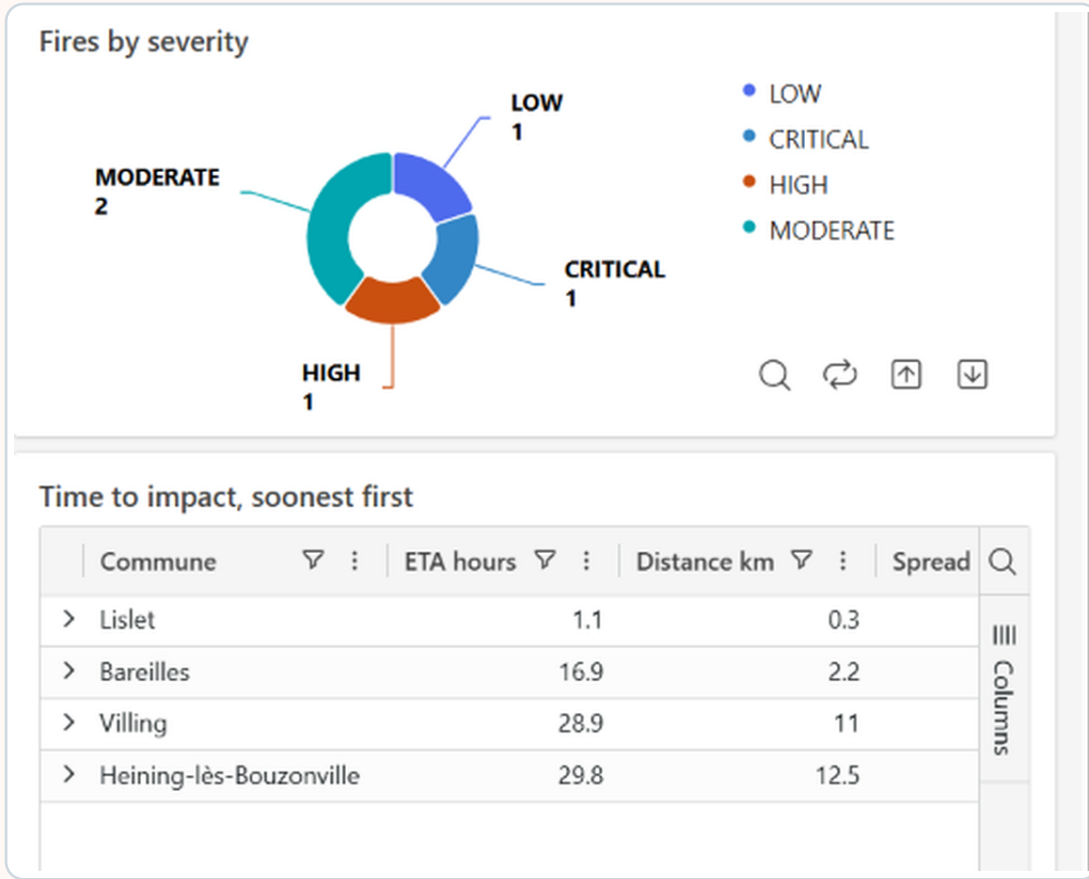
people inside the
exposure footprint

1.1 h

shortest modelled
time to impact

Not all four fires deserve the same attention

Severity is a screening score, and time-to-impact re-orders the queue.



RTD_Wildfire_Command · severity and time to impact

The re-ordering that matters

Heining-lès-Bouzonville

CRITICAL · score 908.7 · GROWING · 201,198 exposed

Villing

HIGH · score 259.1 · stable · 207,966 exposed

Lislet

MODERATE · but ETA 1.1 h — the soonest of all four

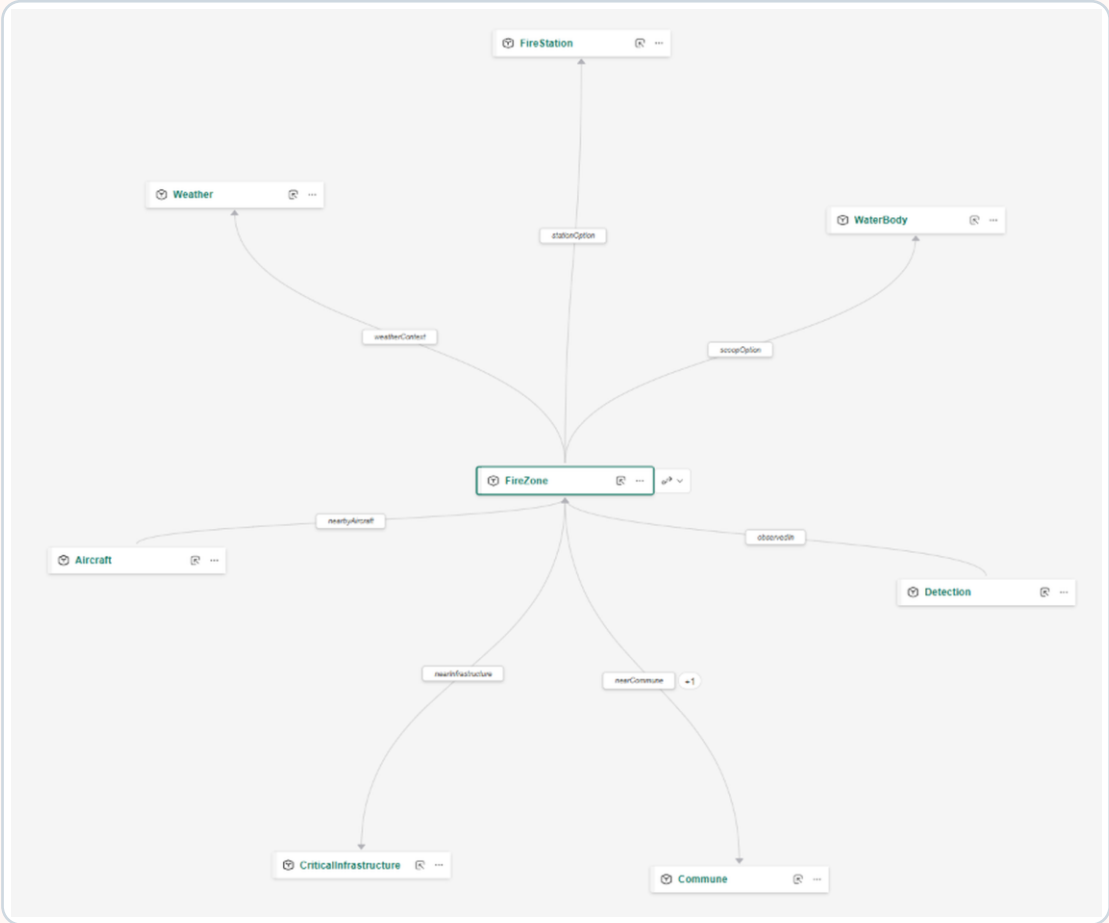
Bareilles

MODERATE · ETA 16.9 h · 2.2 km

The highest score is not always the first job.

Give the data a vocabulary the whole team shares

An ontology turns rows and columns into things a fire chief already has words for.



FROM SCHEMA TO LANGUAGE

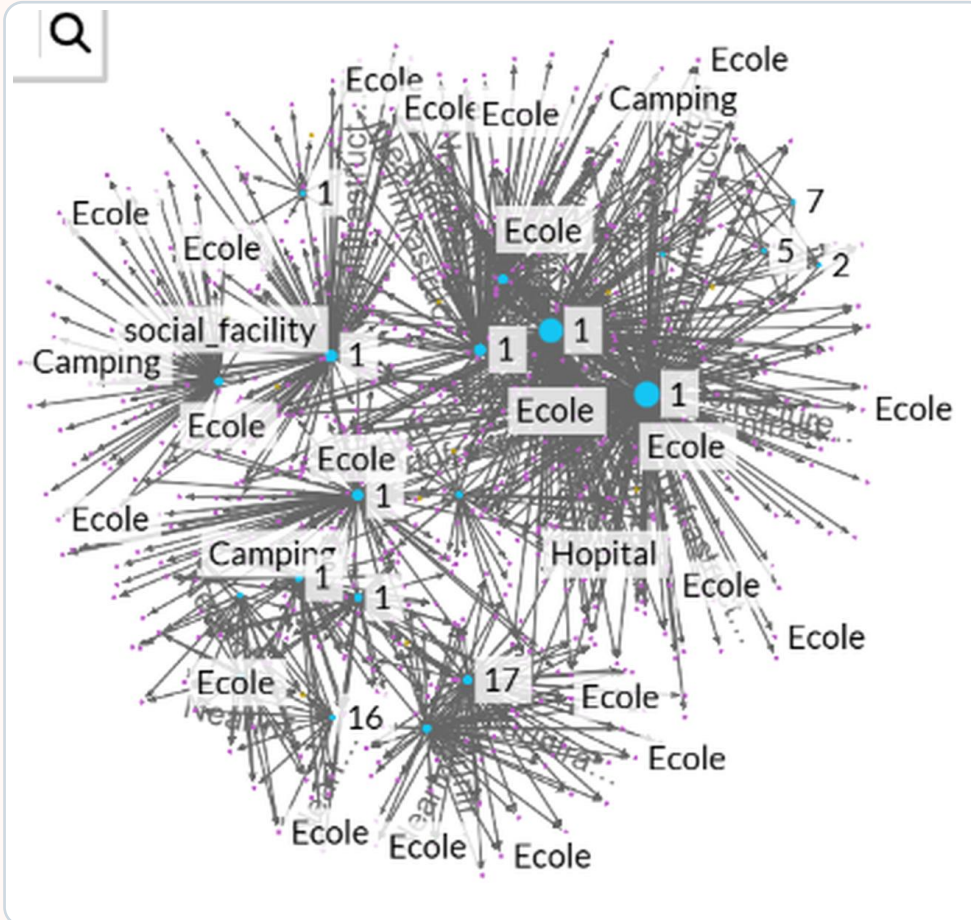
- bronze_fire_raw → **FireZone**
- silver_weather → **Weather**
- ref_communes → **Commune**
- ref_critical_infra → **CriticalInfra**
- ref_fire_stations → **FireStation**

Now an agent — and a human — can ask the same question.

ONT_Wildfire_Impact · 8 entity types, 8 relationships

The map says **WHERE**. The graph says **SO WHAT**.

One fire zone, one traversal — and every school, hospital and camp site attached to it.



THE QUERY

```
MATCH (f:FireZone)
-[:NearCommune]->(c)
-[:NearInfrastructure]->(i)
WHERE f.ObservationStatus
= 'RECENT_IN_SNAPSHOT'
```

That dense burst is not decoration — every spoke is a real École.

Graph analytics where your data resides and processed.

GQS_Wildfire_Story · 00 – Visual impact chain, run live

"Brief me on the evidence."

A Fabric data agent grounded on the ontology — answering in the commander's language.

Which communes are most exposed to fire zones, and what are the nearest fire stations and water bodies for response?

Right now, the most exposed communes are all clustered around the Fos-sur-Mer fire zones in Bouches-du-Rhône (department 13). One persistent hotspot (zones like "12b619f", peak FRP $\approx$ 10 MW, latest frame $\approx$ 5.3 MW, multiple detections) is the main driver.

Operationally, here's the picture, focusing on the larger exposed communes and their nearest resources (from the data returned):

1. Vitrolles (13) – ~36,800 inhabitants

- **Exposure:** Large urban commune within the exposure footprint of the Fos-sur-Mer fire zones (e.g. 12b619f, FRP up to ~10 MW, 12 detections).
- **Nearest fire stations:**
 - CIS Golfe-de-Fos (W707654688) – approx 20–25 km WSW of Vitrolles.
 - CIS Fos-sur-Mer (W104514231) – similar western sector coverage.
- **Nearest scoopable water:**
 - Reservoir "Bassin EP" (W991898209) near Fos-sur-Mer ($\sim$ 4.84°E, 43.46°N), within range for Canadair/Dash scooping on the western flank.

2. Marignane (13) – ~33,700 inhabitants

- **Exposure:** Directly downwind/near the Fos-sur-Mer zones; dense population and critical transport (airport area) make exposure significant.
- **Nearest fire stations:**
 - CIS Golfe-de-Fos (W707654688)

THE PROMPT

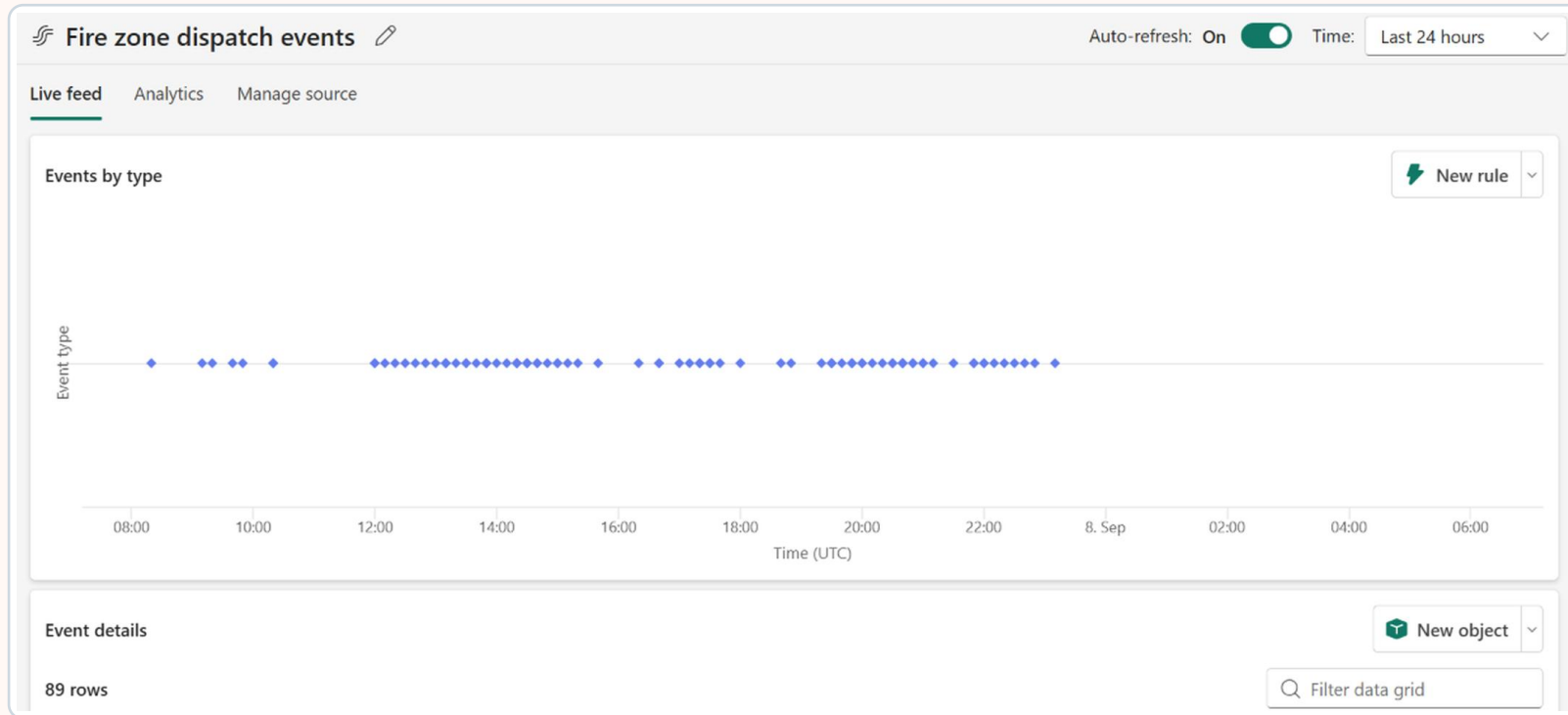
"Which communes are most exposed to fire zones, and what are the nearest fire stations and water bodies for response?"

It named Vitrolles, Marignane, Martigues — with CIS codes.

And it stated its own limits, unprompted.

Attention becomes event-driven — and stays human

Three deliberately narrow rules. Nobody watches a dashboard at 3 a.m.



ACT_Wildfire · 89 real dispatch events in 24 hours

CRITICAL

severity == CRITICAL

GROWING

growth_pct > 40

INTENSITY

single FRP > 50 MW

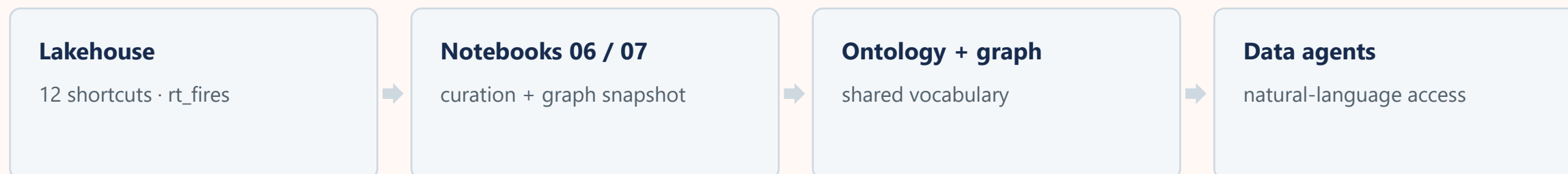
**Three rules, not thirty.
Alert fatigue is a design failure.**

Now here is what was running underneath

One environment. Two serving paths: immediate KQL, and shared Delta context.



OneLake availability · Eventhouse exposes the same tables as Delta — no second pipeline



Medallion, inside one KQL database

Bronze, Silver and Gold are logical layers here — not three more services to buy.

BRONZE | preserve

bronze_fire_raw
bronze_weather_raw
bronze_aircraft_raw

Raw JSON + ingestion time.
Replay and troubleshooting.

SILVER | standardise

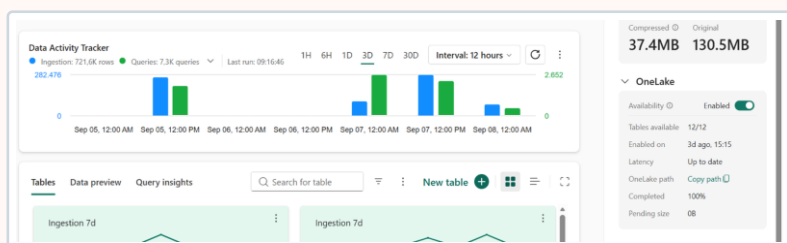
silver_fire_detections
silver_weather
silver_aircraft

Typed columns + S2 spatial keys.
Update policies do the work.

GOLD | serve

gold_fire_dedup
gold_fire_zones
gold_weather_latest

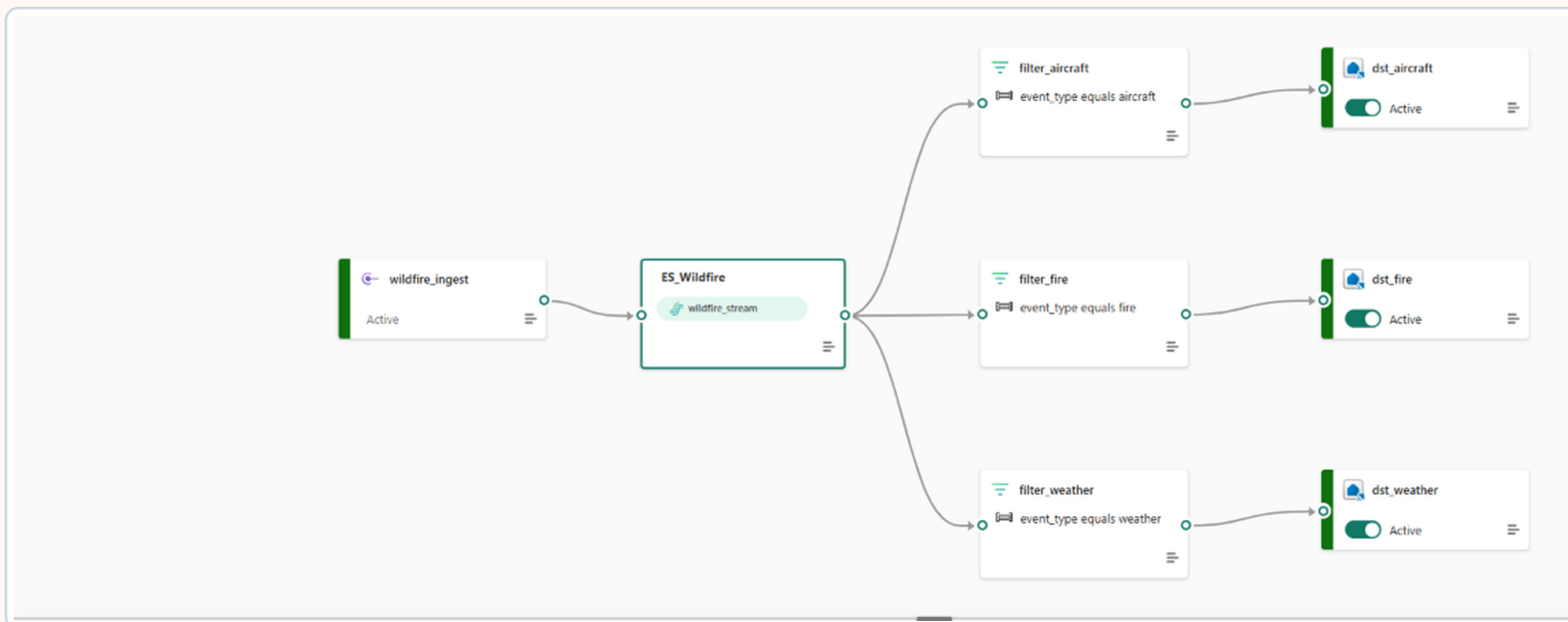
4 materialized views.
Reusable decision-support KQL.



721.6K rows ingested · 7.3K queries served · 130.5 MB compressed to 37.4 MB
One engine. No copy out to a warehouse to make it queryable.

One stream, three deliberate routes

A single CustomEndpoint fans out by event_type. Adding a fourth source is a filter, not a project.



ES_Wildfire · source and all three destinations Active

event_type = fire

bronze_fire_raw

event_type = weather

bronze_weather_raw

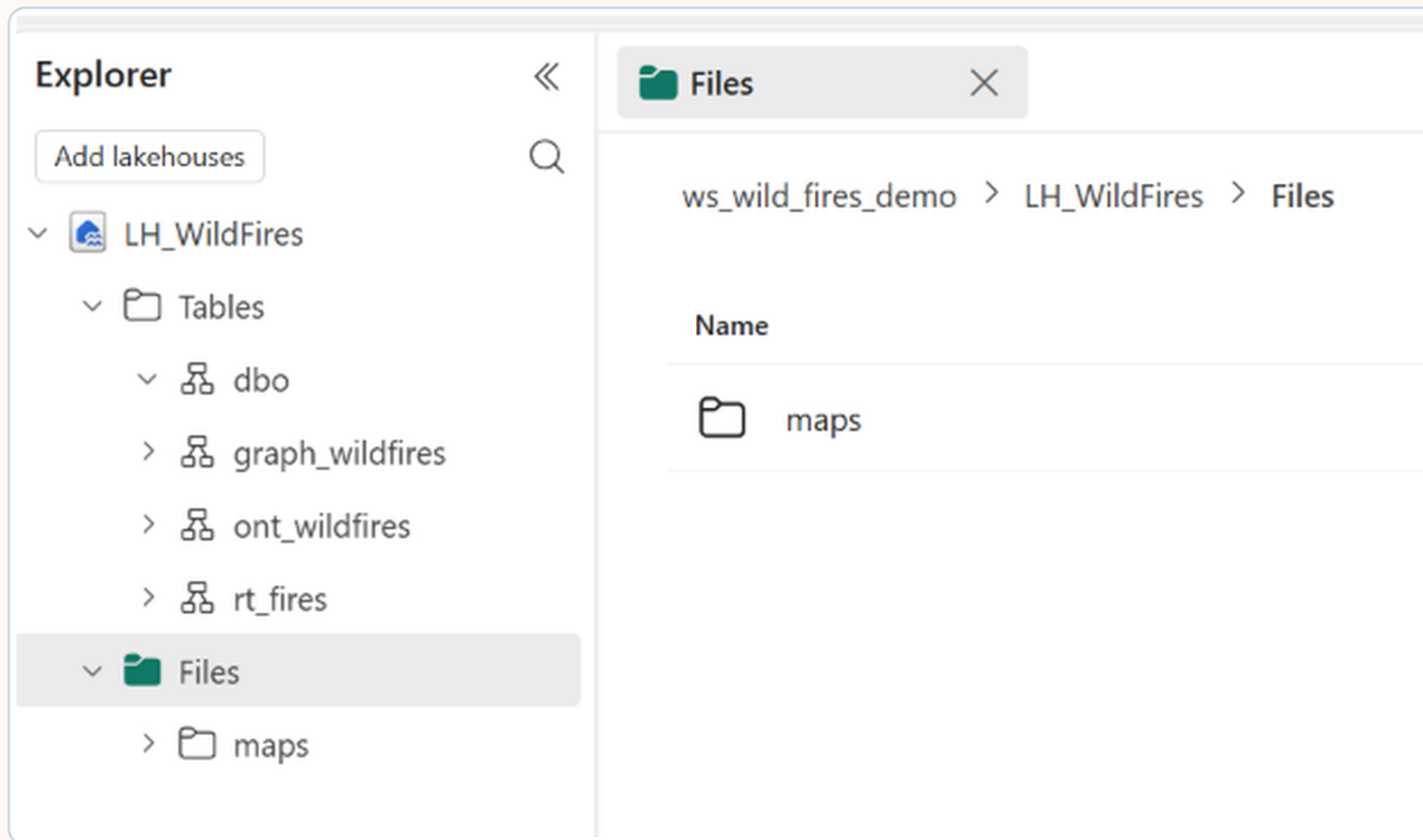
event_type = aircraft

bronze_aircraft_raw

**Watch the destinations.
Active ≠ flowing. Ask me why.**

The part that usually costs you a second pipeline

Eventhouse publishes a Delta representation. Spark, SQL and the ontology all read the same bytes.



Explorer

Add lakehouses

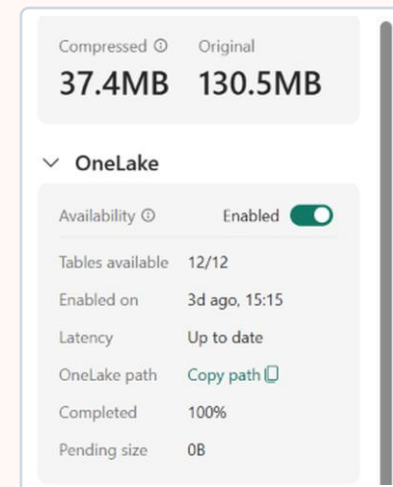
▼ LH_WildFires

- ▼ Tables
 - ▼ dbo
 - > graph_wildfires
 - > ont_wildfires
 - > rt_fires
- ▼ Files
 - > maps

ws_wild_fires_demo > LH_WildFires > Files

Name
maps

LH_WildFires · 3 schemas, 12 OneLake shortcuts



Compressed	Original
37.4MB	130.5MB

▼ OneLake

Availability Enabled

Tables available 12/12

Enabled on 3d ago, 15:15

Latency Up to date

OneLake path [Copy path](#)

Completed 100%

Pending size 0B

WHAT THIS BUYS YOU

- No second ingestion job
- No nightly copy window
- No drift between two versions
- One security boundary

Six engines. One workspace. One story.

Each capability contributes a distinct link in the evidence chain.

Eventstream

Acquire and route
event families

Eventhouse

Query high-volume
time-series evidence

OneLake

Make every table
reusable across engines

Ontology + graph

Name the domain,
traverse relationships

Map + dashboard

Share spatial and
analytical context

Agents + Activator

Ask questions,
route attention

You did not integrate these. They were already integrated.

Credible AI demo is the one that knows its limits

Say this out loud, on stage, before anybody in the room has to ask.

OBSERVED

- Sensor records
- Acquisition + ingestion times
- Reference geography
- Published configuration

INFERRED

- Proximity relationships
- Aircraft role from callsign
- Heuristic priority score
- Downwind screening

NOT ESTABLISHED

- Confirmed fire perimeter
- Actual occupancy or harm
- Verified resource availability
- Validated arrival time

A human validates the situation, the source and the next step before anything operational happens.

This is not really about wildfires

Swap the nouns. The pattern is the same in every operational business you run.

Wildfire

Fire zone → Commune → School

Who do we evacuate first?

Manufacturing

Sensor → Line → Order

Which customer order is at risk?

Energy grid

Fault → Feeder → Hospital

Which critical load drops next?

Logistics

Delay → Hub → Shipment

Which delivery promise breaks?

Real-time signal, plus context, plus a shared vocabulary, plus a human in the loop.

When the wind turns, the question changes.

Fabric connects the evidence fast enough that the whole team
can follow the same answer — and a human still makes the call.

SEE



UNDERSTAND



EXPLAIN



ASK



REVIEW

One observation. One connected picture. A better-informed human.